

MCCLELLANVILLE 7 NE, SC, USA

WMO: 747850

Lat: 33.1532N

Lon: 79.3637W

Elev: 3

StdP: 101.29

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 03728

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.8	-0.9	-11.1	1.4	-0.3	-8.3	1.9	1.7	3.0	15.2	2.9	14.8	0.6	N/A	0.289

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	32.8	25.7	31.7	25.5	30.8	25.0	26.8	31.0	26.3	30.3	25.8	29.5	1.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.7	21.0	29.2	25.2	20.4	28.5	24.8	19.9	28.2	83.8	31.1	81.6	30.4	79.7	29.6	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
2.9	2.4	2.2	DB	-5.5	35.3	2.1	1.2	-7.1	36.2	-8.3	36.9	-9.5	37.6	-11.1	38.5	(4)
			WB	-6.9	28.0	2.2	0.5	-8.5	28.3	-9.8	28.6	-11.0	28.9	-12.6	29.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.5	8.9	10.4	13.7	17.9	22.1	25.7	26.8	26.6	24.4	19.6	13.8	11.2	(6)
		DBStd	7.41	5.43	4.69	4.59	3.69	2.92	2.00	1.55	1.65	2.21	3.90	4.41	4.91	(7)
		HDD10.0	217	88	49	19	1	0	0	0	0	1	14	45		(8)
		HDD18.3	1133	292	224	152	51	6	0	0	1	34	146	227		(9)
		CDD10.0	3311	55	60	134	238	376	471	522	515	433	297	128	81	(10)
		CDD18.3	1186	1	2	9	38	124	221	263	257	183	72	11	4	(11)
		CDH23.3	9421	2	4	34	154	730	1934	2552	2332	1282	363	28	5	(12)
		CDH26.7	2772	0	0	3	16	138	594	884	775	316	46	1	0	(13)
(14)	Wind	WSAvg	0.9	0.9	1.0	1.1	1.2	1.1	1.0	0.9	0.8	0.8	0.8	0.8	0.8	(14)
(15)	Precipitation	PrecAvg	1465	110	82	114	78	83	153	177	191	163	136	71	95	(15)
		PrecMax	1764	256	289	280	204	173	251	277	433	420	298	200	288	(16)
		PrecMin	1042	15	27	29	14	45	63	71	54	23	0	10	29	(17)
		PrecStd	213	69	58	62	51	32	63	62	99	112	90	50	59	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	22.9	24.0	26.3	28.5	32.2	34.0	34.1	33.9	31.7	29.4	25.9	24.0		(19)
		0.4% MCWB	18.5	18.9	17.0	19.9	22.9	25.7	26.2	26.2	24.4	23.6	21.2	20.4		(20)
		2% DB	20.8	21.6	24.2	26.3	29.8	32.3	32.7	32.3	30.5	28.0	24.0	22.1		(21)
		2% MCWB	17.3	16.9	18.2	20.5	22.6	25.4	26.0	25.9	24.3	22.8	19.5	19.4		(22)
		5% DB	18.7	19.6	22.4	25.1	28.0	30.9	31.6	31.3	29.7	26.7	22.5	20.5		(23)
		5% MCWB	16.0	16.1	16.8	19.7	22.4	25.0	25.8	25.5	24.0	22.0	19.0	18.2		(24)
		10% DB	17.1	17.9	20.6	23.9	26.9	29.8	30.6	30.3	28.7	25.5	21.0	18.9		(25)
		10% MCWB	14.7	15.0	16.2	19.0	21.9	24.4	25.3	25.0	23.5	21.4	18.0	17.1		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	19.4	19.9	20.2	23.1	25.0	27.0	27.7	27.6	26.2	25.2	22.0	21.7		(27)
		0.4% MCDB	21.7	23.5	23.8	25.6	29.7	31.9	32.3	31.6	29.3	27.5	24.0	23.3		(28)
		2% WB	18.1	18.5	19.2	22.0	24.0	26.3	26.9	26.8	25.5	24.5	20.7	20.4		(29)
		2% MCDB	20.2	21.1	22.7	25.0	27.5	31.1	31.1	30.6	28.7	26.9	22.4	21.7		(30)
		5% WB	16.8	17.5	18.2	21.2	23.3	25.7	26.4	26.2	25.0	23.5	19.8	18.9		(31)
		5% MCDB	18.4	19.6	21.4	24.2	26.8	29.9	30.4	29.8	28.2	26.0	21.7	20.4		(32)
		10% WB	15.1	16.1	17.3	20.2	22.7	25.2	25.9	25.7	24.5	22.5	18.5	17.5		(33)
		10% MCDB	16.5	18.1	20.0	23.1	26.1	29.0	29.6	29.1	27.6	25.0	20.8	19.1		(34)
(35)	Mean Daily Temperature Range	MDBR	9.7	9.8	9.8	9.6	8.4	7.9	7.7	7.2	7.6	9.0	10.2	9.5		(35)
		5% DB MCDBR	10.3	11.1	11.0	9.7	9.2	9.0	8.6	8.2	8.3	9.0	10.2	9.3		(36)
		5% WB MCWBR	7.4	6.9	5.9	4.9	4.1	4.0	3.9	3.5	3.6	4.7	6.3	7.0		(37)
		5% WB MCDBR	9.2	9.8	9.3	7.9	7.7	8.4	8.1	7.6	7.2	7.4	8.6	8.6		(38)
(40)	Clear-Sky Solar Irradiance	taub	0.331	0.338	0.367	0.403	0.469	0.488	0.532	0.508	0.434	0.393	0.351	0.333		(40)
		taud	2.527	2.507	2.447	2.323	2.195	2.191	2.080	2.143	2.345	2.436	2.508	2.551		(41)
		Ebn at Noon	886	916	911	886	828	809	771	783	832	843	855	864		(42)
		Edh at Noon	84	94	108	128	147	147	163	151	119	101	85	78		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.79	3.45	4.62	5.92	6.42	6.54	6.24	5.60	4.78	4.07	3.10	2.49		(44)
		RadStd	0.17	0.36	0.42	0.41	0.44	0.45	0.38	0.33	0.39	0.44	0.23	0.19		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+256	+160
Regional (41 neighbors)		+0.57	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+194	+137

Nomenclature: See separate page